14b. Staggered Difference-in-Differences

A consolidated, mathematical lecture companion

English notes based on the lecture slides *Staggered Difference-in-Differences* (Causality Lab, Seoul National University, 2025; partly excerpted from B. Callaway's `did` package documentation)

The central question is how to estimate treatment effects with DiD when units adopt treatment at different times and effects may vary across units and over time, a setting in which the usual two-way fixed-effects regression can be badly misleading.

These notes consolidate the slides, explain why TWFE fails, classify valid and invalid comparisons, and develop the Callaway–Sant'Anna group-time ATTs, their identification under two parallel-trends assumptions, and their aggregation. Physical PDF page numbers are listed in Section 3.

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1 Staggered adoption

Purpose. Describe the setting and why it complicates DiD.

The canonical DiD has two periods and two groups. Much applied work has more periods, with units becoming treated at different times: e.g. group A treated early, group B later, group C later still, and a never-treated group. The slides' figures (staggered DiD without and with treatment reversal) show outcome paths of groups A, B, C over 15 periods: after adoption the treated paths jump and may keep changing (dynamic effects) relative to their counterfactual (triangles); with treatment reversal, effects switch on and off.

Staggered DiD generalises DiD to settings where treatment timing varies, common when policies are rolled out gradually across regions: a *staggered adoption design* (SAD). Units not yet treated can serve as controls for those already treated, so staggered adoption blurs the distinctions treatment/control and pre/post. Once treated, a unit remains treated (no reversal).

1.1 TWFE as is is not sufficient

The most common approach is the TWFE regression $Y_{it} = \alpha_i + \beta_t + \tau D_{it} + \varepsilon_{it}$. It works when (i) effects are not heterogeneous, or (ii) there are only two periods: then, under parallel trends and no anticipation, τ equals the ATT. But effects are very likely heterogeneous, varying across units or over time. With staggered timing, already-treated units exhibit treatment dynamics and act as controls; these dynamics leak into τ, making it very hard to interpret causally. The fix: use only "good" comparisons between groups instead of all possible comparisons.

1.2 Terminology

- **Static effects:** constant across post-treatment periods (e.g. a persistent 5% increase).
- **Dynamic effects:** change with time since adoption (grow, decay, fluctuate).
- **Homogeneous effects:** identical across units.
- **Heterogeneous effects:** vary across units (group- or unit-level differences) but are constant over time for each unit.
- **Carryover effects:** a past treatment keeps influencing outcomes after it is withdrawn; a specific form of dynamic effect.

1.3 Valid and invalid comparisons

At time t, a treated unit can be compared with never-treated units, not-yet-treated (later-treated) units, or already-treated (earlier-treated) units.

- **Valid:** treated vs. never-treated; treated vs. not-yet-treated.
- **Invalid:** treated vs. already-treated; later-treated vs. earlier-treated using the earlier group's post-treatment periods.

Bias arises when effects vary over time (dynamic) or across units (heterogeneous). (The Goodman-Bacon decomposition in Lecture 14 shows TWFE averages all 2×2 comparisons, including the invalid ones.)

Modern alternatives to TWFE modify the set of effective comparison units: Callaway and Sant'Anna (2021), group-time ATT estimators; Sun and Abraham (2021), event-study designs

with proper controls; stacked DiD, separate DiDs per cohort to avoid contamination. They differ in which observations serve as comparisons and how covariates enter.

2 Callaway–Sant'Anna group-time ATTs

Purpose. Define, identify and aggregate effects by adoption cohort and calendar time.

2.1 Notation

G_i: the period in which unit i becomes treated (groups are defined by adoption time). $Y_{it}(0)$: untreated potential outcome in period t. $Y_{it}(g)$: potential outcome in period t if first treated in period g. Y_{it}: observed outcome. Never-treated units have $Y_{it} = Y_{it}(0)$ for all t; for others

$$Y_{it} = \mathbb{1}\{G_i > t\}Y_{it}(0) + \mathbb{1}\{G_i \leq t\}Y_{it}(G_i) :$$

untreated outcomes before participation and treated outcomes (possibly depending on adoption time) afterwards. This notation embeds a no-anticipation assumption, which can be relaxed. C_i: indicator of the never-treated (control) group. D_{it}: indicator that i has been treated by t. X_i: pre-treatment covariates.

Staggered adoption assumption: for $t = 1, \ldots, T-1$, $D_{it} = 1 \Rightarrow D_{i,t+1} = 1$: once treated, units remain treated (they do not "forget" their treatment). Effects may be heterogeneous across time, groups, treatment sequences, etc.

2.2 Parallel trends

Based on never-treated units: for all $g = 2, \ldots, T$ and $t = 2, \ldots, T$ with $t \geq g$,

$$\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid G = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid C = 1].$$

It uses the never-treated for all eventually-treated groups, presuming (i) a large enough never-treated group exists and (ii) these units are similar enough.

Based on not-yet-treated units: for all g and s, t with $s \geq t \geq g$,

$$\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid G = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid D_s = 0, G \neq g].$$

Units not yet treated by time $s \geq t$ can serve as comparisons when computing effects for the group first treated at g. This generally uses more data, but restricts some pre-treatment trends across groups (Marcus and Sant'Anna 2021): no free lunch. (The slides ask, as an exercise, why the restriction is placed on s relative to t; note that $s \geq t$ guarantees, by staggered adoption, that comparison units are untreated in both periods $t-1$ and t.)

2.3 Group-time ATT and identification

$$\mathrm{ATT}(g, t) = \mathbb{E}[Y_t(g) - Y_t(0) \mid G = g],$$

the average effect for the group first treated in g, at time t. With three periods, $\mathrm{ATT}(g = 2, t = 3)$ is the effect in period 3 for units first treated in period 2.

Under either parallel-trends assumption, group-time ATTs are identified. With never-treated controls, for $t \geq g$,

$$\mathrm{ATT}(g, t) = \mathbb{E}[Y_t - Y_{g-1} \mid G = g] - \mathbb{E}[Y_t - Y_{g-1} \mid C = 1];$$

with not-yet-treated controls, for $t \geq g$,

$$\mathrm{ATT}(g,t) = \mathbb{E}[Y_t - Y_{g-1} \mid G = g] - \mathbb{E}[Y_t - Y_{g-1} \mid D_t = 0, G \neq g].$$

Derivation (never-treated case): $\mathbb{E}[Y_t(g) \mid G = g] = \mathbb{E}[Y_t \mid G = g]$ by the observation rule. Telescoping parallel trends from g to t, $\mathbb{E}[Y_t(0) - Y_{g-1}(0) \mid G = g] = \mathbb{E}[Y_t(0) - Y_{g-1}(0) \mid C = 1] = \mathbb{E}[Y_t - Y_{g-1} \mid C = 1]$; and $Y_{g-1}(0) = Y_{g-1}$ for group g (no anticipation). Combine. Each $\mathrm{ATT}(g,t)$ is a clean 2×2 DiD between group g and a valid comparison group, with base period $g - 1$.

2.4 Conditional parallel trends

Parallel trends are often more plausible conditional on pre-treatment covariates. *Never-treated:* for $t \geq g \geq 2$, $\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, G = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, C = 1]$. *Not-yet-treated:* for $s \geq t \geq g \geq 2$, $\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, G = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, D_s = 0, G \neq g]$. These allow covariate-specific trends, important when covariate distributions differ across groups (estimated by outcome regression, IPW or doubly robust DiD, as in Lecture 14).

2.5 Aggregation

By group. The average effect of participation for group g:

$$\theta_S(g) = \frac{1}{T - g + 1} \sum_{t=2}^{T} \mathbb{1}\{g \leq t\} \, \mathrm{ATT}(g,t),$$

highlighting heterogeneity with respect to adoption time. Aggregating further,

$$\theta_S^O := \sum_{g=2}^{T} \theta_S(g) \, P(G = g)$$

is the overall effect of participating across all groups that ever participate: close to a multi-period analogue of the two-period ATT, and the recommended single summary parameter.

By exposure length (event study). To highlight dynamics,

$$\theta_D(e) := \sum_{g=2}^{T} \mathbb{1}\{g + e \leq T\} \, \mathrm{ATT}(g, g+e) \, P(G = g \mid g + e \leq T),$$

the average effect for units exposed to treatment for exactly e periods.

2.6 Summary

- Staggered DiD uses group-time ATTs $\mathrm{ATT}(g,t)$: the effect in period t for units first treated in period g.
- It allows flexible clean controls: not-yet-treated or never-treated units.
- It accommodates pre-treatment (static) covariates via conditional parallel trends.
- Anticipation can be handled by shifting the base period before any behavioural adjustment.

References: Athey and Imbens (2022); Baker, Larcker and Wang (2022); Callaway and Sant'Anna (2021); Chiu, Lan, Liu and Xu (2023); Goodman-Bacon (2021); Wing et al. (2024).

3 Source coverage

Topic	Physical pages
Staggered adoption, figures, definition	1–6
TWFE failure, terminology, comparisons, alternatives	7–11
Notation and assumptions	12–16
Group-time ATT, identification, conditional PT, aggregation, summary	17–23